

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Exponential and Logarithmic Functions

Week 2 Day 2 PM



SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

OUTLINE

① Laws of Logarithms

② Exponential and Logarithmic Equations

③ Fitting Exponential and Power Curves to Data

④ Summary

LAWS OF LOGARITHMS

The laws of logarithms can be derived from the following defining property of inverse functions:

$$a^y = x \Leftrightarrow \log_a(x) = y.$$

- For $x, y > 0$, since $a^{\log_a(x) + \log_a(y)} = a^{\log_a(x)} \cdot a^{\log_a(y)} = xy$, thus

$$\log_a(xy) = \log_a(x) + \log_a(y).$$

- For $x, y > 0$, since $a^{\log_a(x) - \log_a(y)} = \frac{a^{\log_a(x)}}{a^{\log_a(y)}} = \frac{x}{y}$, thus

$$\log_a\left(\frac{x}{y}\right) = \log_a(x) - \log_a(y).$$

- For $x > 0$, since $a^{y \log_a(x)} = \left(a^{\log_a(x)}\right)^y = x^y$, thus

$$\log_a(x^y) = y \log_a(x).$$

LAWS OF LOGARITHMS

⚠ There is **no** corresponding rule for logarithm of a sum or a difference.



$$\log_a(x + y) \neq \log_a(x) + \log_a(y).$$



$$\log_a(x + y) \neq \log_a(x) \cdot \log_a(y).$$



$$\log_a(x - y) \neq \log_a(x) - \log_a(y).$$



$$\log_a(x - y) \neq \frac{\log_a(x)}{\log_a(y)}.$$

LAWS OF LOGARITHMS - EXAMPLE

E.g.: $\log_4 2 + \log_4 32 = \log_4(2 \times 32) = \log_4 64 = \log_4 4^3 = 3.$

E.g.: $\log_2 80 - \log_2 5 = \log_2 \left(\frac{80}{5}\right) = \log_2 16 = \log_2 2^4 = 4.$

E.g.: $-\frac{1}{3} \ln 8 = \ln \left(8^{-\frac{1}{3}}\right) = \ln \left(\frac{1}{8^{\frac{1}{3}}}\right) = \ln \frac{1}{2}.$

E.g.: $\log_2(6x) = \log_2 6 + \log_2 x = \log_2(2 \cdot 3) + \log_2 x =$
 $\log_2 2 + \log_2 3 + \log_2 x = 1 + \log_2 3 + \log_2 x.$

E.g.: For $x, y > 0$, $\lg(x^3 y^6) = \lg x^3 + \lg y^6 = 3 \lg x + 6 \lg y.$

E.g.: $\ln(-2)^2 \neq 2 \ln(-2)$, because $\log_a(x^y) = y \log_a(x)$ holds for $x > 0$ only. After all, $\ln(-2)$ is undefined. In general, if n is an even integer, then $\log_a x^n = \log_a |x|^n = n \log_a |x|$ for all $x \in \mathbb{R}$.

E.g.: For $a, b, c > 0$,

$$\ln \left(\frac{ab}{\sqrt[3]{c}}\right) = \ln a + \ln b - \ln c^{\frac{1}{3}} = \ln a + \ln b - \frac{1}{3} \ln c.$$

ACTIVITY 1

(a) Evaluate the following expressions without using a calculator.

(i) $\log_2 60 - \log_2 15$

(ii) $\log_5 \frac{1}{\sqrt{125}}$

(iii) $\ln \left(\ln e^{e^{2024}} \right)$

(b) Expand the following expressions, assuming $x, y > 0$.

(i) $\log_3(x\sqrt{y})$

(ii) $\lg \sqrt{x^2 + y^2}$

(iii) $\ln \sqrt{x\sqrt{y}}$

(c) Combine the following expressions.

(i) $2 \log_2 x - \frac{1}{3} \log_2(x + 1)$

(ii) $4 \lg x - \lg(x^2 - 1) + 2 \lg(x - 1)$

ACTIVITY 1: SOLUTION

$$(a) \quad (i) \log_2 60 - \log_2 15 = \log_2 \frac{60}{15} = \log_2 4 = 2.$$

$$(ii) \log_5 \frac{1}{\sqrt{125}} = \log_5 \frac{1}{\sqrt{5^3}} = \log_5 5^{-\frac{3}{2}} = -\frac{3}{2}.$$

$$(iii) \ln \left(\ln e^{2024} \right) = \ln e^{2024} = 2024.$$

$$(b) \quad (i) \log_3(x\sqrt{y}) = \log_3 x + \frac{1}{2} \log_3 y.$$

$$(ii) \lg \sqrt[2]{x^2 + y^2} = \frac{1}{2} \lg(x^2 + y^2).$$

$$(iii) \ln \sqrt{x\sqrt{y}} = \frac{1}{2} \ln(x\sqrt{y}) = \frac{1}{2} \ln x + \frac{1}{2} \ln y^{\frac{1}{2}} = \\ \frac{1}{2} \ln x + \frac{1}{4} \ln y.$$

$$(c) \quad (i) 2 \log_2 x - \frac{1}{3} \log_2(x+1) = \log_2 \frac{x^2}{\sqrt[3]{x+1}}.$$

$$(ii) 4 \lg x - \lg(x^2 - 1) + 2 \lg(x - 1) = \lg \frac{x^4(x-1)^2}{x^2-1} = \\ \lg \frac{x^4(x-1)^2}{(x+1)(x-1)} = \lg \frac{x^4(x-1)}{x+1}.$$

CHANGE OF BASE FORMULA

CHANGE OF BASE FORMULA

Let $a, b > 0$ be two numbers $\neq 1$. Then for any $x > 0$, we have

$$\log_b(x) = \frac{\log_a x}{\log_a b}.$$

Proof: Let $y = \log_b(x)$, then

$$b^y = x$$

$$\Leftrightarrow \log_a b^y = \log_a x$$

$$\Leftrightarrow y \log_a b = \log_a x$$

$$\Leftrightarrow \log_b(x) = y = \frac{\log_a x}{\log_a b}. \quad \square$$

E.g.: $\log_3 7 = \frac{\log_2 7}{\log_2 3}.$

E.g.: $\log_5 10 = \frac{\log_{10} 10}{\log_{10} 5} = \frac{1}{\lg 5}.$

ACTIVITY 2

- (a) Rewrite the following in terms of $\ln$ and then evaluate them using a calculator.
- (i) $\log_2 5$
 - (ii) $\lg 8787174$
 - (iii) $\log_{\pi} e$
- (b) By using the change of base formula, describe the transformation needed to graph $y = \log_3 x$ from the graph $y = \ln x$.

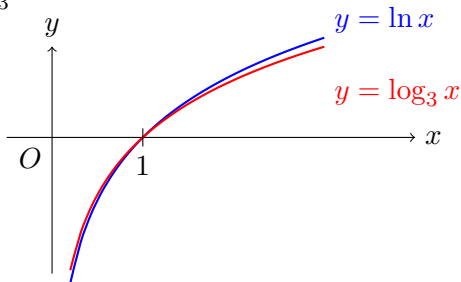
ACTIVITY 2: SOLUTION

(a) (i) $\log_2 5 = \frac{\ln 5}{\ln 2} = 2.3219 \dots$

(ii) $\lg 8787174 = \frac{\ln 8787174}{\ln 10} = 6.9438 \dots$

(iii) $\log_\pi e = \frac{\ln e}{\ln \pi} = 0.8735 \dots$

- (b) Since $y = \log_3 x = \frac{\ln x}{\ln 3} = \frac{1}{\ln 3} \ln x$, the graph of $y = \log_3 x$ is obtained by shrinking the graph of $y = \ln x$ vertically by a factor of $\frac{1}{\ln 3}$.



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- ① Laws of Logarithms
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EXPONENTIAL EQUATIONS

An **exponential equation** is one in which the variable occurs in the exponent. The general strategy to solve an exponential equation is to isolate the exponential expression on one side of the equation and then apply logarithm to each side to “bring down the exponent”. After that, just solve for the variable as usual.

E.g.: To solve $3^{x+2} = 7$, we apply $\ln$ to both sides and get

$$\ln 3^{x+2} = \ln 7 \Leftrightarrow (x+2) \ln 3 = \ln 7 \Leftrightarrow x = \frac{\ln 7}{\ln 3} - 2.$$

E.g.: To solve $4(1 + 10^{5x}) = 9$, we first isolate 10^{5x} on one side:

$$4(1 + 10^{5x}) = 9 \Leftrightarrow 1 + 10^{5x} = \frac{9}{4} \Leftrightarrow 10^{5x} = \frac{9}{4} - 1 = \frac{5}{4}.$$

Now, we apply $\ln$ on both sides:

$$\ln 10^{5x} = \ln \frac{5}{4} \Leftrightarrow 5x \ln 10 = \ln \frac{5}{4} \Leftrightarrow x = \frac{1}{5 \ln 10} \ln \frac{5}{4}.$$

SOLVING EXPONENTIAL EQUATIONS BY SUBSTITUTION

If a^{2x} and a^x are present in an exponential equation, we can try substituting $y = a^x$ to convert it into a quadratic equation.

E.g.: To solve $9^x + 3^{x+1} = 10$, we first simplify it to

$$(3^2)^x + 3 \cdot 3^x - 10 = 0 \Leftrightarrow (3^x)^2 + 3 \cdot 3^x - 10 = 0.$$

By substituting $y = 3^x$, we get

$$y^2 + 3y - 10 = 0 \Leftrightarrow (y + 5)(y - 2) = 0 \Leftrightarrow y = -5 \text{ or } 2.$$

However, since $y = 3^x > 0$, so the only solution is

$$3^x = 2 \Leftrightarrow \ln 3^x = \ln 2 \Leftrightarrow x \ln 3 = \ln 2 \Leftrightarrow x = \frac{\ln 2}{\ln 3}.$$

SOLVING EXPONENTIAL EQUATIONS BY DIVIDING a^x

If a^x appears as a factor for the entire equation, then we can divide both sides by a^x , as $a^x \neq 0$, so the division is valid.

E.g.: To solve $3xe^x + x^2e^x = 0$, we first factorize it to

$$(3x + x^2)e^x = 0.$$

By dividing e^x , we get

$$3x + x^2 = 0 \Leftrightarrow x(3 + x) = 0 \Leftrightarrow x = -3 \text{ or } 0.$$

ACTIVITY 3

Solve the following equations.

(a) $7^{\frac{x}{2}} = 5^{1-x}$

(b) $e^{2x} - 3e^x + 2 = 0$

(c) $x^2 2^x = 3x^4 2^{x+1}$

ACTIVITY 3: SOLUTION

(a)

$$7^{\frac{x}{2}} = 5^{1-x} \quad \Leftrightarrow \quad \frac{x}{2} \ln 7 = (1-x) \ln 5$$

$$\Leftrightarrow \frac{x}{2} \ln 7 + x \ln 5 = \ln 5 \quad \Leftrightarrow \quad x \left(\frac{1}{2} \ln 7 + \ln 5 \right) = \ln 5$$

$$\Leftrightarrow x = \frac{\ln 5}{\frac{1}{2} \ln 7 + \ln 5} = \frac{\ln 5}{\ln \sqrt{7} + \ln 5} = \frac{\ln 5}{\ln(5\sqrt{7})}.$$

ACTIVITY 3: SOLUTION

(b) By substituting $y = e^x$, we have

$$\begin{aligned} e^{2x} - 3e^x + 2 &= 0 &\Leftrightarrow &y^2 - 3y + 2 = 0 \\ \Leftrightarrow (y - 2)(y - 1) &= 0 &\Leftrightarrow &e^x = y = 1 \text{ or } 2 \\ &&\Leftrightarrow &x = \ln 1 = 0 \text{ or } \ln 2. \end{aligned}$$

(c)

$$\begin{aligned} x^2 2^x &= 3x^4 2^{x+1} &\Leftrightarrow &x^2 2^x - 3x^4 2^{x+1} = 0 \\ \Leftrightarrow 2^x x^2 (1 - 2 \cdot 3x^2) &= 0 &\Leftrightarrow &x^2 (1 - 6x^2) = 0. \\ &&\Leftrightarrow &x = 0 \text{ or } \pm \frac{1}{\sqrt{6}}. \end{aligned}$$

LOGARITHMIC EQUATIONS

A **logarithmic equation** is one in which a logarithm of the variable occurs. The general strategy to solve a logarithmic equation is to isolate the logarithmic term on one side of the equation and then raise the base to the power of each side to remove the logarithm. After that, just solve for the variable as usual.

E.g.: To solve $\log_2(25 - x) = 3$, we raise 2 to the power to each side and get

$$25 - x = 2^3 \Leftrightarrow x = 25 - 8 = 17.$$

E.g.: To solve $\lg(x + 2) + \lg(x - 1) = 1$, we first combine the logarithmic terms and then raise 10 to the power to each side:

$$\begin{aligned}\lg(x + 2) + \lg(x - 1) &= 1 && \Leftrightarrow \lg((x + 2)(x - 1)) = 1 \\ \Leftrightarrow x^2 + x - 2 &= 10^1 && \Leftrightarrow x^2 + x - 12 = 0 \\ \Leftrightarrow (x + 4)(x - 3) &= 0 && \Rightarrow x = 3.\end{aligned}$$

We reject $x = -4$ because $\lg(-4 + 2)$ and $\lg(-4 - 1)$ are undefined.

ACTIVITY 4

Solve the following equations.

(a) $\ln x + \ln(x - 1) = \ln(4x)$

(b) $4 - \log_2(3 - x) = \log_4 3$

(c) $\ln x^2 - \ln x = 5$

ACTIVITY 4: SOLUTION

(a)

$$\begin{aligned}\ln x + \ln(x - 1) &= \ln(4x) &\Leftrightarrow &\ln(x(x - 1)) = \ln(4x) \\ \Rightarrow &x(x - 1) = 4x &\Leftrightarrow &x^2 - x - 4x = 0 \\ \Leftrightarrow &x(x - 5) = 0 &\Leftrightarrow &x = 0 \text{ or } 5.\end{aligned}$$

However, for $\ln x$, $\ln(x - 1)$ and $\ln(4x)$ to be defined, we must have $x > 0$ and $x > 1$, i.e. $x > 1$. So the only solution is $x = 5$.

ACTIVITY 4: SOLUTION

(b)

$$4 - \log_2(3 - x) = \log_4 3 \quad \Leftrightarrow \quad 4 - \log_2(3 - x) = \frac{\log_2 3}{\log_2 4}$$

$$\Leftrightarrow \quad 4 - \log_2(3 - x) = \frac{\log_2 3}{2} \quad \Leftrightarrow \quad 8 - 2\log_2(3 - x) = \log_2 3$$

$$\Leftrightarrow \quad \log_2 3 + 2\log_2(3 - x) = 8 \quad \Rightarrow \quad \log_2(3(3 - x)^2) = 8$$

$$\Leftrightarrow \quad 3(3 - x)^2 = 2^8 \quad \Leftrightarrow \quad 3 - x = \pm \frac{2^4}{\sqrt{3}}$$
$$\Leftrightarrow \quad x = 3 \mp \frac{2^4}{\sqrt{3}}.$$

However, for $\log_2(3 - x)$ to be defined, we must have $x < 3$.

So the only solution is $x = 3 - \frac{2^4}{\sqrt{3}}$.

ACTIVITY 4: SOLUTION

(c)

$$\ln x^2 - \ln x = 5 \quad \Leftrightarrow \quad \ln \frac{x^2}{x} = 5$$

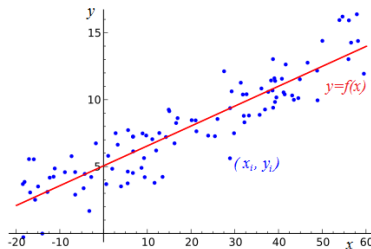
$$\Leftrightarrow \quad x = e^5.$$

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LINEAR REGRESSION

Suppose we want to study the relationship between variables y and x . We can collect data points (x_i, y_i) and try to find a function $y = f(x)$ that fits the data best.



This is the general idea of **regression** techniques: to model the relationship between variables based on empirical data. If the data points scatter close to a line (like the one shown above), then it's reasonable to model the relationship as

$$y = mx + c, \quad \text{where } m, c \text{ are constants.}$$

This is called **linear regression**.

LINEAR REGRESSION

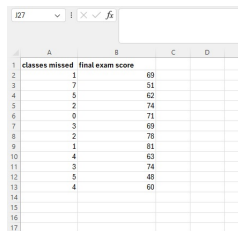
In order to find the line $y = mx + c$ that “fits the data the best” (called the **regression line**), we need to find the m and c such that the “error” between the model and the data is minimized, which can be done using multivariable differential calculus. At this point, we will simply use Excel to find the regression line. We’ll illustrate it using an example.

E.g.: The following table shows the number of classes missed (x) and the final exam scores (y) of 12 students. Let’s find the regression line using Excel.

no. of classes missed	final exam score
1	69
7	51
5	62
2	74
0	71
3	69
2	78
1	81
4	63
3	74
5	48
4	60

LINEAR REGRESSION IN EXCEL - EXAMPLE

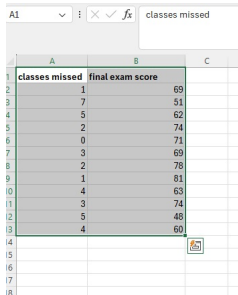
1. Copy the data into two columns in an Excel sheet.



A screenshot of an Excel spreadsheet. The formula bar at the top shows 'I27' and 'fx'. The spreadsheet has columns A, B, C, D, and E. Column A is labeled 'classes missed' and column B is labeled 'final exam score'. The data is as follows:

	A	B	C	D	E
1	classes missed	final exam score			
2	1	69			
3	7	51			
4	5	62			
5	2	74			
6	0	71			
7	3	69			
8	2	78			
9	1	81			
10	4	63			
11	3	74			
12	5	48			
13	4	60			
14					
15					
16					
17					

2. Left click and hold to select the data.

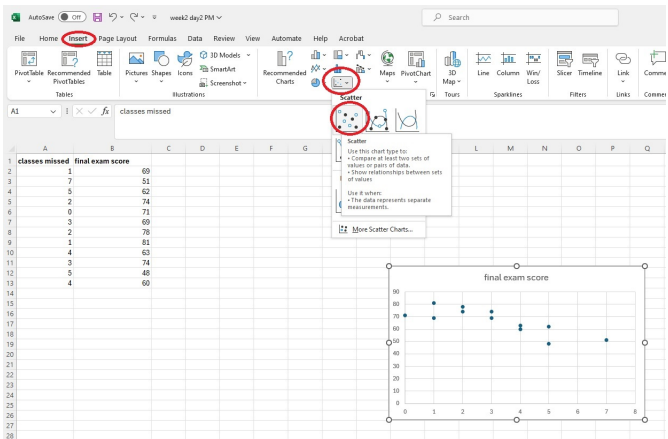


A screenshot of an Excel spreadsheet showing the same data as the previous table. The data range from A1 to B13 is selected, indicated by a green border. The formula bar at the top shows 'A1' and 'classes missed'.

	A	B	C
1	classes missed	final exam score	
2	1	69	
3	7	51	
4	5	62	
5	2	74	
6	0	71	
7	3	69	
8	2	78	
9	1	81	
10	4	63	
11	3	74	
12	5	48	
13	4	60	
14			
15			
16			
17			
18			

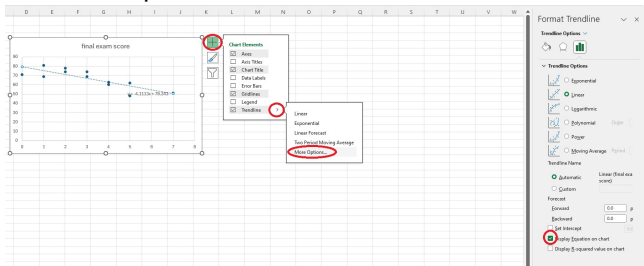
LINEAR REGRESSION IN EXCEL - EXAMPLE

3. Go to Insert and then select the scatter plot in the charts section.



LINEAR REGRESSION IN EXCEL - EXAMPLE

4. Left click the plus sign on the top right corner, move your cursor to Trendline and click on the > symbol next to it, then click on More Options...

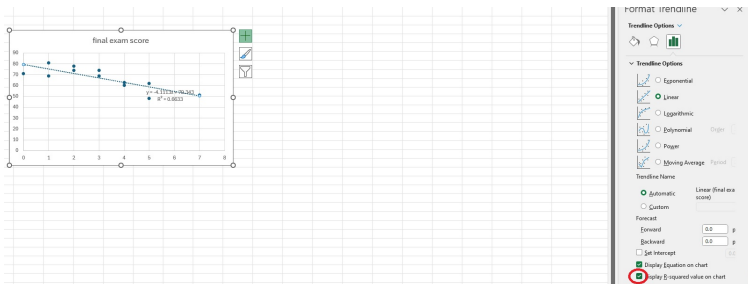


5. The Format Trendline menu will appear on the right. The linear option is selected by default. To display the equation of the regression line, select Display Equation on chart at the bottom. In this example, we get

$$y = -4.1113x + 79.343.$$

LINEAR REGRESSION IN EXCEL - EXAMPLE

6. You can get the ***coefficient of determination***/ R^2 ***value*** by selecting Display R-squared value on chart. It's a value between 0 and 1 which tells us how good the fitting is. 0 means there's no linear relationship between x and y , whereas 1 means a perfect linear relationship between x and y .



$R^2 = 0.6633$ for our data, which means the linear relationship is not very strong.

MODELLING WITH EXPONENTIAL FUNCTIONS

If the scatter plot of a set of data (x_i, y_i) resembles a exponential graph, then we might want to model the data as

$$y = Ce^{kx}, \quad \text{where } k, C \text{ are constants.}$$

In order to find the optimal C, k such that the model $y = Ce^{kx}$ fits the data the best, we can linearize the data and then apply linear regression to find C and k . More precisely,

$$y = Ce^{kx} \Leftrightarrow \ln y = kx + \ln C.$$

Thus, if we define new data $Y := \ln y$ and $X := x$, then we expect the data (X_i, Y_i) to scatter close to the regression line

$$Y = kX + \ln C.$$

SEMILOG PLOT

So from the table of x_i, y_i , we calculate a new table of $X_i = x_i, Y_i = \ln y_i$ and then apply Excel to get the regression line for X, Y :

$$Y = mX + c.$$

From here, we substitute back $Y = \ln y$ and $X = x$ to find the exponential relationship between x and y :

$$Y = mX + c$$

$$\Leftrightarrow \ln y = mx + c$$

$$\Leftrightarrow y = e^{mx+c}$$

$$\Leftrightarrow y = e^c \cdot e^{mx}.$$

Note that if the scatter plot of $(x_i, \ln y_i)$ (called the **semilog plot**) doesn't resemble a line, then it strongly suggests that the exponential model is not suitable.

ACTIVITY 5

The table below gives the population of the world in the 20th century.

Year (t)	World population (P in millions)
1900	1650
1910	1750
1920	1860
1930	2070
1940	2300
1950	2520
1960	3020
1970	3700
1980	4450
1990	5300
2000	6060

- (a) Make a scatter plot of (t, P) . Is a linear model appropriate?
- (b) Construct a new table with $X = t$ and $Y = \ln P$ as columns in Excel.
- (c) Make a scatter plot of (X, Y) . Is a linear model appropriate?
- (d) Use Excel to find the regression line of X, Y . What's the R^2 value? Is this regression line a good fitting?
- (e) Hence, find the regression model for t, P .

ACTIVITY 5: SOLUTION

- (a) See the Excel file. No, the plot doesn't resemble a line.
- (b) See the Excel file.
- (c) See the Excel file. Yes, the plot resembles a line.
- (d) We get $Y = 0.0136X - 18.613$. The R^2 value is 0.9697, which means the fitting is very good.
- (e)

$$Y = 0.0136X - 18.613$$

$$\Leftrightarrow \ln P = 0.0136t - 18.613$$

$$\Leftrightarrow P = e^{0.0136t - 18.613}$$

$$\Leftrightarrow y = e^{-18.613} \cdot e^{0.0136t} = 8.25 \times 10^{-9} e^{0.0136t}.$$

MODELLING WITH POWER FUNCTIONS

If the scatter plot of a set of data (x_i, y_i) resembles the graph of a power function, then we might want to model the data as

$$y = ax^n, \quad \text{where } a, n \text{ are constants.}$$

In order to find the optimal a, n such that the model $y = ax^n$ fits the data the best, we can linearize the data and then apply linear regression to find a and n . More precisely,

$$y = ax^n \Leftrightarrow \ln y = n \ln x + \ln a.$$

Thus, if we define new data $Y := \ln y$ and $X := \ln x$, then we expect the data (X_i, Y_i) to scatter close to the regression line

$$Y = nX + \ln a.$$

LOG-LOG PLOT

So from the table of x_i, y_i , we calculate a new table of $X_i = \ln x_i, Y_i = \ln y_i$ and then apply Excel to get the regression line for X, Y :

$$Y = mX + c.$$

From here, we substitute back $Y = \ln y$ and $X = \ln x$ to find the exponential relationship between x and y :

$$Y = mX + c$$

$$\Leftrightarrow \ln y = m \ln x + c$$

$$\Leftrightarrow y = e^{m \ln x + c}$$

$$\Leftrightarrow y = e^c \cdot x^m.$$

Note that if the scatter plot of $(\ln x_i, \ln y_i)$ (called the **log-log plot**) doesn't resemble a line, then it strongly suggests that the power function model is not suitable.

ACTIVITY 6 [DO IT AT HOME]

The table below gives the mean distance d of each planet from the sun in astronomical units and its period T in years.

Planet	d	T
Mercury	0.387	0.241
Venus	0.723	0.615
Earth	1.000	1.000
Mars	1.523	1.881
Jupiter	5.203	11.861
Saturn	9.541	29.457
Uranus	19.190	84.008
Neptune	30.086	164.784
Pluto	39.507	248.350

- Make a scatter plot of (d, T) . Is a linear model appropriate?
- Construct a new table with $X = \ln d$ and $Y = \ln T$ as columns in Excel.
- Make a scatter plot of (X, Y) . Is a linear model appropriate?
- Use Excel to find the regression line of X, Y . What's the R^2 value? Is this regression line a good fitting?
- Hence, find the regression model for d, T .

ACTIVITY 6: SOLUTION

- (a) See the Excel file. No, the plot doesn't resemble a line.
- (b) See the Excel file.
- (c) See the Excel file. Yes, the plot resembles a line.
- (d) We get $Y = 1.4997X + 0.0004$. The R^2 value is 0.9999999592, which means the fitting is very very good.
- (e)

$$Y = 1.4997X + 0.0004$$

$$\Leftrightarrow \ln T = 1.4997 \ln d + 0.0004$$

$$\Leftrightarrow T = e^{1.4997 \ln d + 0.0004}$$

$$\Leftrightarrow T = e^{0.0004} \cdot d^{1.4997} = 1.00040008d^{1.4997} \approx \sqrt{d^3}.$$

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SUMMARY

- Laws of Logarithms
 - laws of logarithms for products, quotients and powers.
 - the change of base formula.
- Exponential and Logarithmic Equations
 - solving exponential equations by isolating the exponential.
 - solving exponential equations by substitution.
 - solving exponential equations by dividing the common exponential factor.
 - solving logarithmic equations.
- Fitting Exponential and Power Curves to Data
 - linear regression.
 - linear regression in Excel.
 - modelling with exponential functions and semilog plot.
 - modelling with power functions and log-log plot.

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 4.4, 4.5. Sections 4.6, 4.7 are not covered in this course, but it's good to know.